Experiment

# A1-1

English (Official)

# Fiber Bragg grating

## Part A. FBG fabrication technology (5.0 points)

**A.1** (0.2pt)

$\Lambda =$

**A.2** (0.4pt)

$v_0 =$

**A.3** (0.2pt)

$\lambda_B =$

**A.4** (0.2pt)

$n_c =$

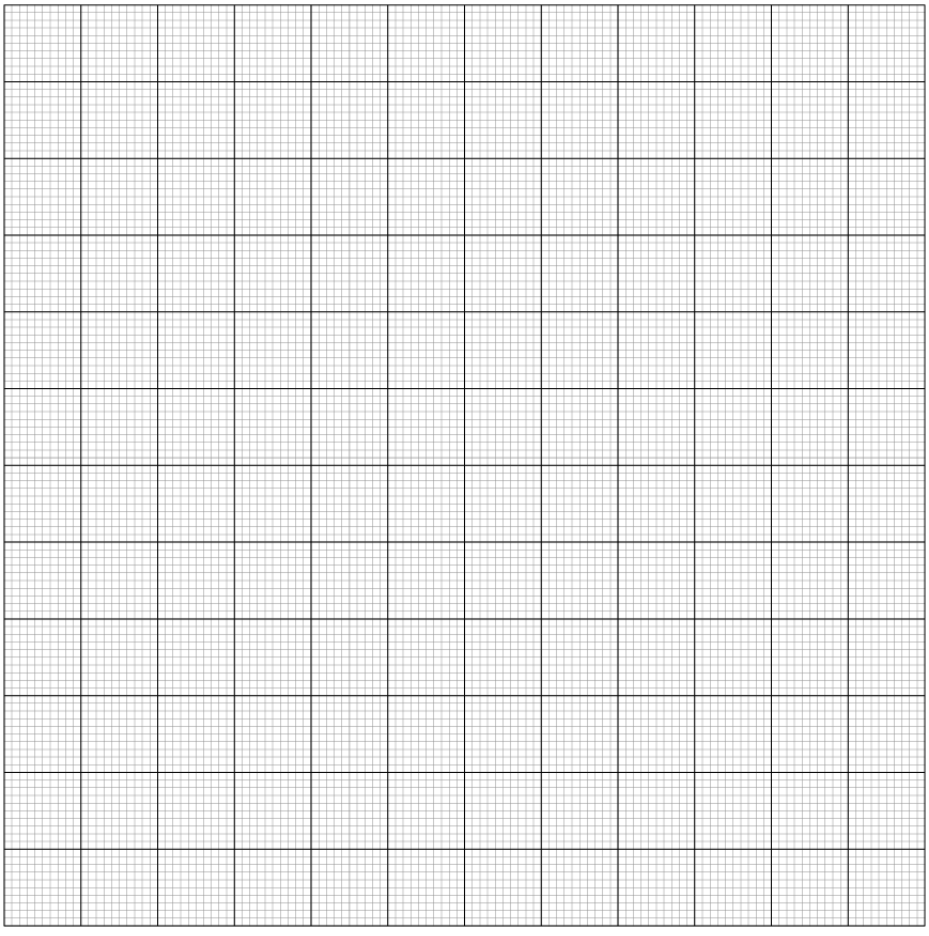
**A.5** (0.2pt)

$v_{1550} =$

**A.6** (1.8pt)

| $\Delta Y$ ($\mu$m) | | | | | | | |
|---|---|---|---|---|---|---|---|
| $T_{min}$ (dB) | | | | | | | |

| $\Delta Y$ ($\mu$m) | | | | | | | |
|---|---|---|---|---|---|---|---|
| $T_{min}$ (dB) | | | | | | | |



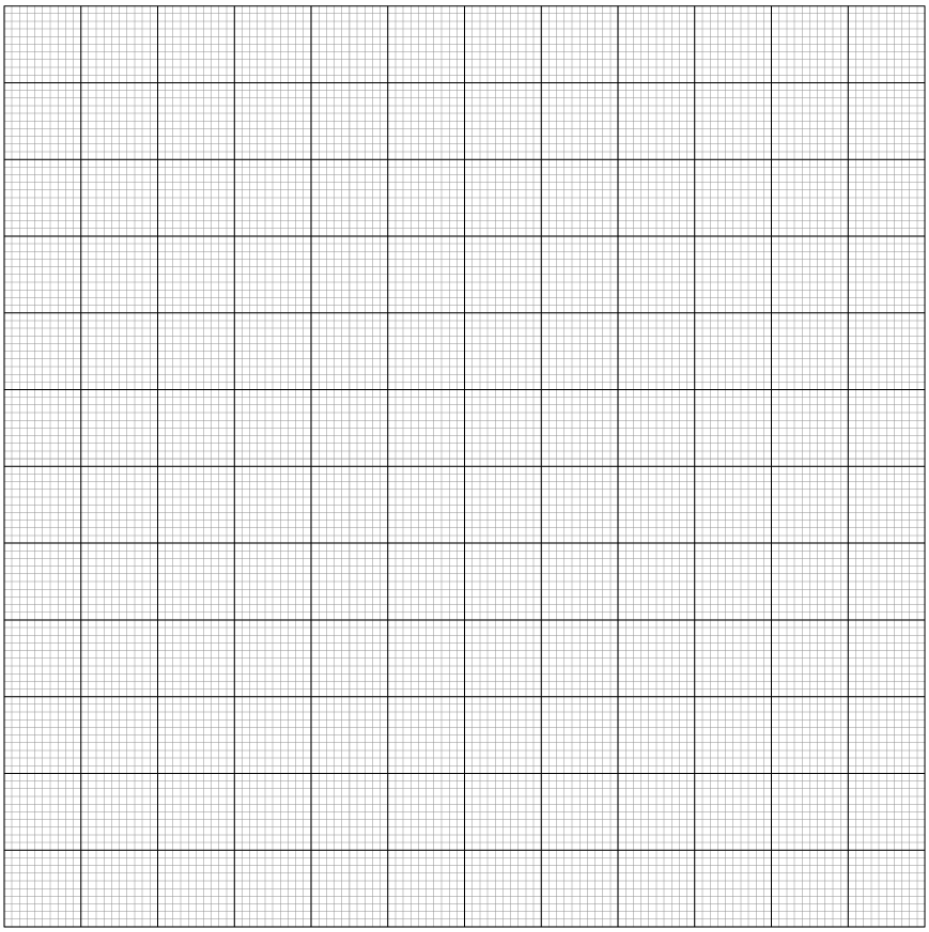
**A.7** (0.3pt)

$\Delta Y_0 =$

**A.8** (1.7pt)

| $L$ (mm) | | | | | | |
|---|---|---|---|---|---|---|
| $T_{min}$ (dB) | | | | | | |

| $L$ (mm) | | | | | | |
|---|---|---|---|---|---|---|
| $T_{min}$ (dB) | | | | | | |



$k =$

## Part B. Coupled mode theory (2.9 points)

**B.1** (0.5pt)

**B.2** (1.4pt)

$T =$

**B.3** (1.0pt)

$\varkappa_0 =$

## Part C. Calibration of an FBG temperature sensor (2.9 points)

**C.1** (0.4pt)

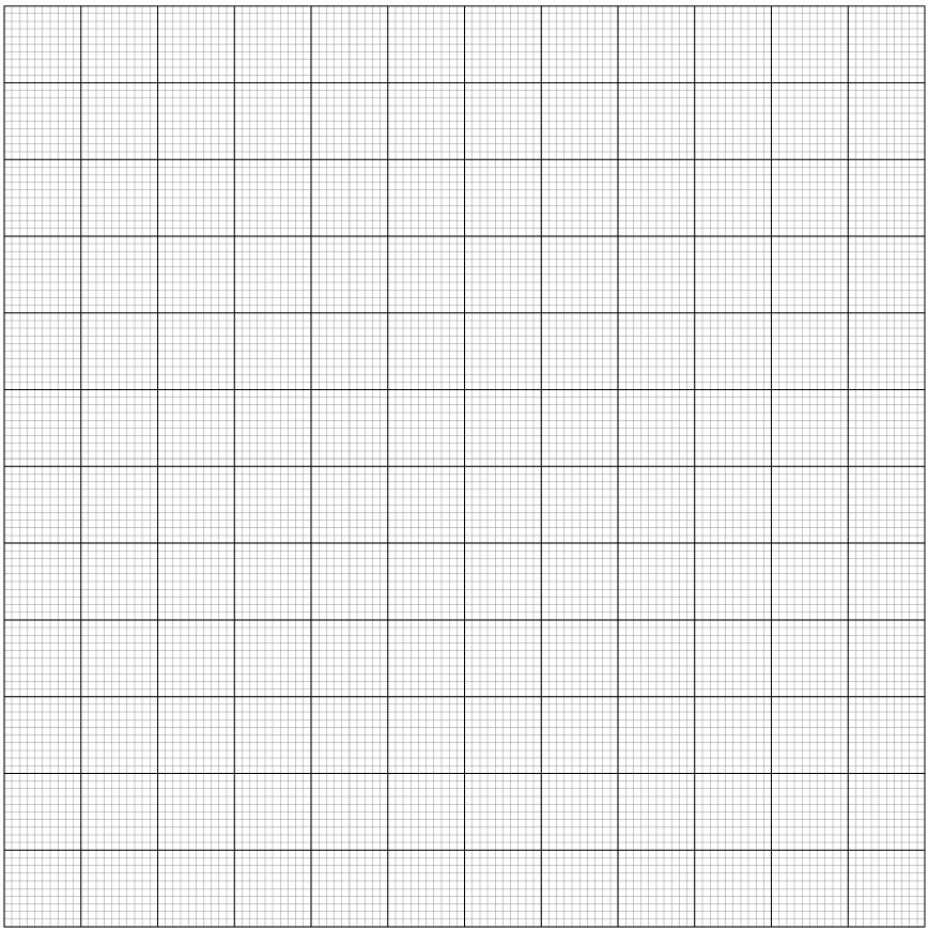
$A =$

**C.2** (1.2pt)

| | | | | | | |
|---|---|---|---|---|---|---|
| $t$ (°C) | | | | | | |
| $\lambda_B$ (nm) | | | | | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| $t$ (°C) | | | | | | |
| $\lambda_B$ (nm) | | | | | | |

**C.3** (0.8pt)



$A =$

**C.4** (0.5pt)

## Part D. Using the FBG temperature sensor (2.4 points)

**D.1** (1.6pt)

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| $\tau$ (min) | | | | | | | | |
| $\lambda_B$ (nm) | | | | | | | | |
| $t$ (°C) | | | | | | | | |

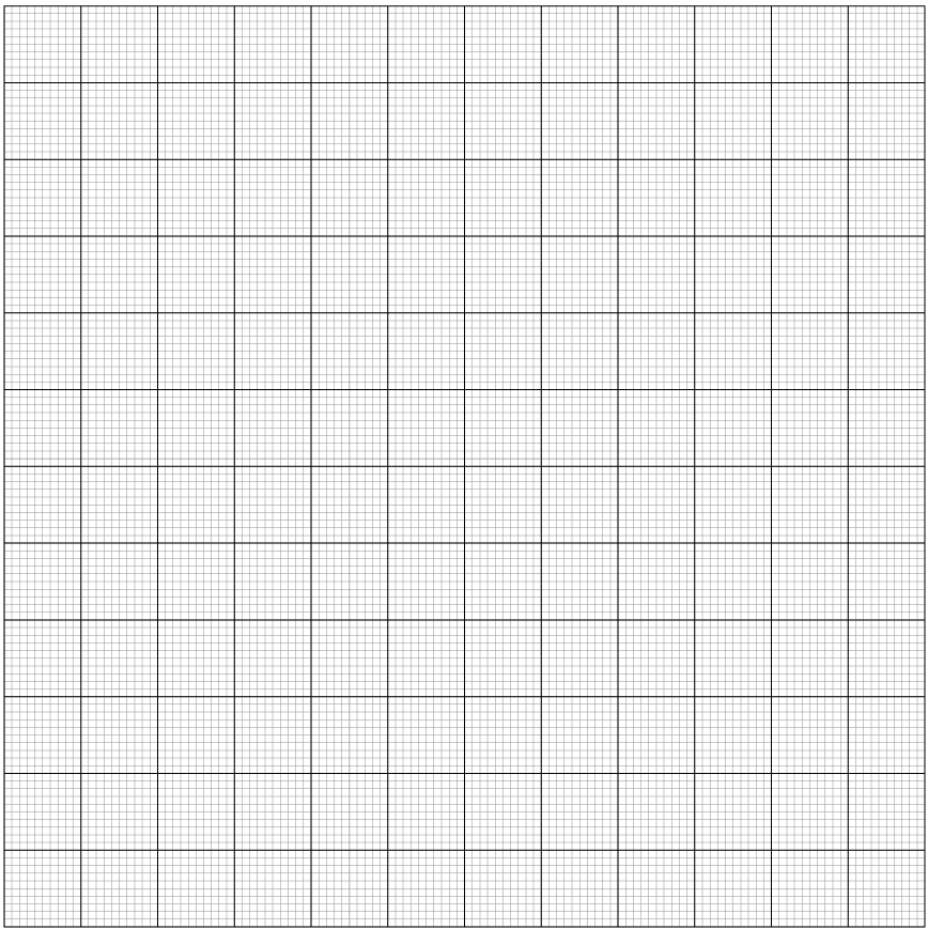
| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| $\tau$ (min) | | | | | | | | |
| $\lambda_B$ (nm) | | | | | | | | |
| $t$ (°C) | | | | | | | | |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| $\tau$ (min) | | | | | | | | |
| $\lambda_B$ (nm) | | | | | | | | |
| $t$ (°C) | | | | | | | | |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| $\tau$ (min) | | | | | | | | |
| $\lambda_B$ (nm) | | | | | | | | |
| $t$ (°C) | | | | | | | | |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| $\tau$ (min) | | | | | | | | |
| $\lambda_B$ (nm) | | | | | | | | |
| $t$ (°C) | | | | | | | | |

**D.2** (0.8pt)



$t_0 =$

## Part E. FBG dynamometer (3.9 points)

**E.1** (0.5pt)

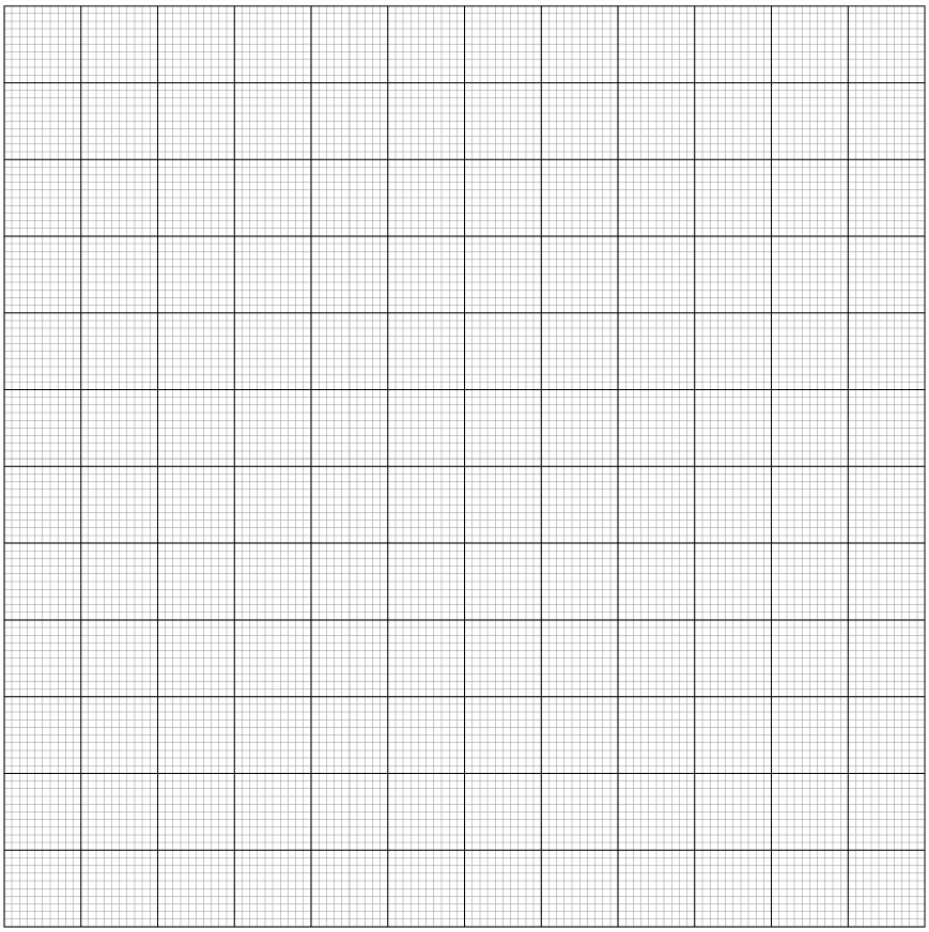
$B =$

**E.2** (1.0pt)

| $F$ (mN) | | | | | | |
|---|---|---|---|---|---|---|
| $\lambda_B$ (nm) | | | | | | |

| $F$ (mN) | | | | | | |
|---|---|---|---|---|---|---|
| $\lambda_B$ (nm) | | | | | | |

**E.3** (0.7pt)

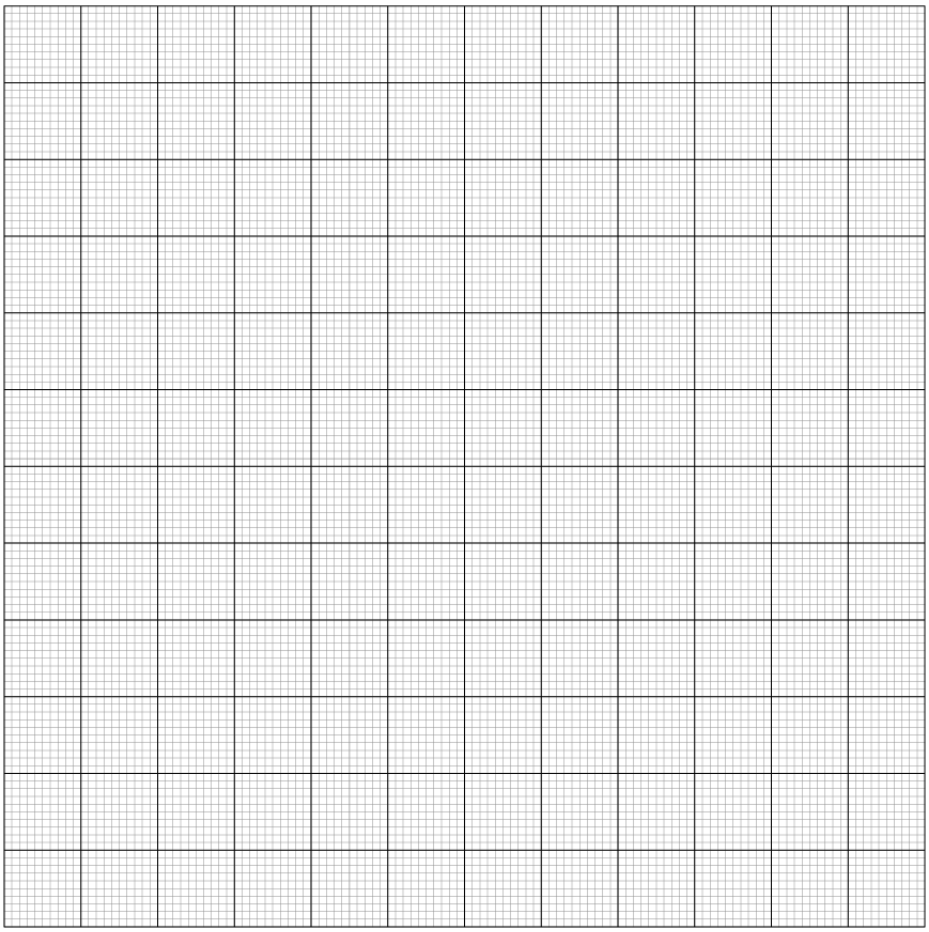


$B =$

**E.4** (1.7pt)

| $\Delta x$ (mm) | | | | | | |
|---|---|---|---|---|---|---|
| $\lambda_B$ (nm) | | | | | | |

| $\Delta x$ (mm) | | | | | | |
|---|---|---|---|---|---|---|
| $\lambda_B$ (nm) | | | | | | |



$k =$

## Part F. Birefringence in FBG (2.9 points)

**F.1** (2.0pt)

| | | | | | | |
|---|---|---|---|---|---|---|
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|---|---|---|---|---|---|---|
| | | | | | | |

$\theta_0 =$

**F.2** (0.9pt)

$\varkappa_{\parallel} =$ $\varkappa_{\perp} =$

$\lambda_{B,\parallel} =$ $\lambda_{B,\perp} =$